



A CDER perspective: Landscape of New Approach Methodologies (NAMs) submitted in drug development programs

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ARTICLE INFO

Handling Editor: Dr. Daniele Wikoff

Keywords:

New approach methodologies (NAMs)
Drug development
Nonclinical testing
FDA
CDER
Regulatory toxicology

ABSTRACT

This study analyzes the landscape of studies using New Approach Methodologies (NAMs) and submitted to the FDA over the past 15 years. The study utilized a custom-developed Center for Drug Evaluation and Research (CDER) search tool to examine keywords in Module 4 of the Electronic Common Technical Document (eCTD) across various drug applications. The investigation focused on five NAM categories, including some of those identified in the 2022 Food and Drug Omnibus Reform Act (FDORA): in vitro, in silico, in chemico, nonhuman in vivo, and other NAMs. Results indicated that 93 % of NAM submissions were concentrated in two categories: in silico (49 %) and in vitro (44 %), with other categories (including in chemico, nonhuman in vivo from phylogenetically lower species and other combined methods) demonstrating significantly lower representation. In vitro NAMs, including stem cell-derived and sandwich culture models, showed higher prevalence compared to 3D models and organ chip or MPS models. The relatively high prevalence of in silico NAMs, was due to the submission of multiple study reports per application, attributed to various metabolites and the use of different in silico platforms. In chemico and nonhuman in vivo NAMs demonstrated limited submissions, with zebrafish studies predominating in the latter category. This study highlights one of CDER's activities aimed at better understanding the current usage of NAMs in drug development, while providing evidence to support areas of focus and prioritization of resources towards the validation of NAMs with significant regulatory impact potential. Despite technical limitations in the datamining work presented here, the findings confirm that CDER has been receiving NAMs data in drug applications for a long time. Nevertheless, we acknowledge that industry submits only a fraction of their NAM data to FDA, therefore we encourage increased NAM submissions which would contribute to building scientific confidence in these methods. It is only through the availability of sufficient case studies, that CDER can move towards reaching the goals of NAMs Roadmap issued in April 2025, and ultimately phasing out animal studies when possible and feasible.

1. Introduction

The Food and Drug Omnibus Reform Act of 2022 (FDORA) updated section 701 to the Federal Food, Drug, and Cosmetic Act (FD&C Act) which amended the language to remove “animal” and “pre-clinical tests (including tests on animals)” for small molecule drugs and biosimilars and replacing it with “nonclinical tests”. FDORA added language to define nonclinical tests as “a test conducted in vitro, in silico, or in chemico, or a nonhuman in vivo test, that occurs before or during the clinical trial phase of the investigation of the safety and effectiveness of a drug”. Such test may include the following:

1. Cell-based assays.
2. Organ chips and microphysiological systems.
3. Computer modeling.
4. Other nonhuman or human biology-based test methods.
5. Animal tests.

To facilitate a discussion about specific NAMs it is crucial to harmonize definitions for commonly used keywords and concepts. In the absence of a consensus definition, we concur with the definition of NAMs previous stated in FDA publications (Avila et al., 2020). However, for the present work, we have separated the terms into categories using the FDORA language definition for nonclinical tests. Therefore, we used

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<https://doi.org/10.1016/j.yrtph.2025.106007>

Received 26 September 2025; Received in revised form 13 November 2025; Accepted 29 November 2025

Available online 6 December 2025

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the FDORA language mentioned above, as a guide to help us organize the wide range of applications, tools and techniques into five broad categories (Table 1). These categories are not intended to be formal definitions, but rather a general distinction between different types of NAMs, only for the purposes of helping us in our datamining of CDER databases. A more detailed table of commonly used definitions and keywords related to NAMs is provided in Appendix A.

- 1. in vitro** – refers to the technique of performing a given procedure in a controlled environment outside of a living organism. This usually involves studying isolated cells by themselves or in concert with others.
Examples include:
 - o Cell based assays (these include 2D cell culture systems)
 - Primary cells (coculture, triculture)
 - Stem cell derived (pluripotent and multipotent stem cells)
 - Sandwich culture
 - o 3D in vitro
 - Organoids, spheroid, assembloids
 - Reconstructed human tissue (such as epidermis and cornea)
 - o Microphysiological systems (MPS)
 - Organ chips
- 2. in silico (or computational toxicology)** – approaches that attempt to understand and predict toxicity mechanisms and potential toxic effects using mathematics, informatics, and computer models
- 3. in chemico** – represents abiotic chemical reactivity methods, usually physicochemical measurements that do not contain biological material (with the exception of certain tests that employ enzyme fractions)
- 4. nonhuman in vivo test (in vivo)** – refers to experimentation using a whole, living organism
- 5. Other/Combined Approaches** – for situations that either do not easily fit into one of the categories above or combine aspects of multiple categories

To understand the current landscape of NAMs submitted to FDA, we conducted a comprehensive analysis of FDA databases spanning the last 15 years. Our search focused on the categories and examples of NAMs as defined by FDORA. This analysis aimed to provide insights into the types and frequency of NAMs being submitted to the FDA in support of safety assessment in drug development. This analysis does not assess the maturity, qualification, or validation status of the assays proposed for inclusion in regulatory decision-making processes. It's important to note that our data analysis does not include assays and technologies which are beyond the scope outlined in FDORA, such as ex vivo models, humanized mouse models, microarrays, and receptor screening assays. Our analysis allowed us to align our findings more closely with the current regulatory framework thus providing a clearer picture of NAMs adoption in FDA submissions over the past 15 years.

Table 1
General categories of keywords used to search FDA database.

Category of NAMs	Examples
In vitro	3D models (organoids, spheroids), organ chip and microfluidic systems, coculture, sandwich culture, stem cell derived models, reconstructed tissue models
In silico	Quantitative structure activity (QSAR), high throughput screen, predictive model, read across
In chemico	Direct peptide reactivity assay
Nonhuman In vivo	Alternative organisms, such as <i>C. elegans</i> , zebrafish
Other	bioprinting, bioengineering, Multi-electrode array (MEA), TEER (Trans Epithelial Electrical Resistance)

2. Methods

An internal CDER search tool was used to search for keywords contained in Module 4 of the Electronic Common Technical Document (eCTD), which contains nonclinical study reports for various drug applications, including Investigational New Drug (IND) applications, New Drug Applications (NDAs), and Biologics License Applications (BLAs). Our keyword selection process involved gathering over 130 keywords from multiple sources to generate a robust and unbiased database of terms related to NAMs: internal FDA experience, collaboration with industry (under a Private Public Partnership (PPP) with IQ Microphysiological Systems Affiliate (IQ MPS)) and published literature. The keywords were refined through an iterative process, by combining related keywords, condensing keywords into root words and then grouping keywords under the FDORA categories (described above) for efficient data mining. The search tool was designed to search the title of nonclinical study reports submitted in eCDT. Data was collected from each keyword for IND, NDA and BLA applications over a 15-year period, January 2010–December 2024.

3. Results

Based on the categories highlighted in FDORA we have listed specific examples of NAMs (Table 1). A comprehensive list of keywords used in our searches is detailed in the respective sections below. It's important to note that relevant to the total number of submissions (IND, NDA, BLA) the prevalence of individual NAM types is relatively low. This analysis provides insight into the current industry habits of submitting NAMs in regulatory submissions, highlighting both the dominance of certain methodologies and the overall low frequency of specific other NAM categories.

The most common NAMs found in CDER submissions are predominantly concentrated in two categories: in silico (49 % of total NAM submissions) and in vitro (44 % of total NAM submissions) (Fig. 1). However, these total numbers of in silico and in vitro submissions are limited, at only 0.28 % and 0.25 % respectively, when compared to the total amount of study reports submitted in IND, NDA and BLA submissions in the past 15 years (data not shown). These two NAM categories combined represent 93 % of all NAM submissions, using the keywords that we selected for this analysis. The remaining categories show significantly lower representation: these include other NAMs (4 % of total NAM submissions), nonhuman in vivo methods (3 % of total NAM submissions), in chemico methods (0.8 % of total NAM submissions) (Fig. 1). Notably, in chemico methods have the lowest representation among all NAM categories.

In Table 2, we have listed the specific keywords used in our search under the in vitro category, as well as how many INDs and BLAs or NDAs contain those keywords in their study titles. A variety of in vitro NAMs have been submitted to CDER including from the most to least common: stem cell derived models (n = 499), sandwich culture models (n = 252), coculture models (n = 220), 3D models (n = 131), reconstructed tissue models (n = 63) and organ chip or MPS models (n = 13) (Table 2, Fig. 2a). It's important to note that some categories, such as stem cell derived models, 3D models, and organ chip/MPS models, are broader and encompass multiple keywords (Fig. 2b). An analysis of submission trends revealed that significantly more in vitro NAMs were submitted in IND applications compared to NDA and BLA applications (Fig. 2c).

In silico NAMs were generally detected with two main keywords, “in silico” (66 %) or “quantitative structure activity OR QSAR” (33 %), with fewer hits for other related keywords (1 %) such as “computational method OR computational model” or “high throughput screen” (Table 3, Fig. 3a). The data shown represent the total number of study reports submitted annually that include either “in silico” or “quantitative structure activity or QSAR” in the study report title. However, it's

important to note that these figures may be somewhat skewed due to potential duplications, study reports that contain both the keywords “in silico” or “quantitative structure activity OR QSAR” in the study title, leading to double counting. We found that when combining “quantitative structure activity” OR “QSAR” and “in silico” keywords, there were 105 study reports from 2010 to 2024, which contained both keywords in the title. However, even with duplicates, it does not significantly alter results in Fig. 1 since the most prominent NAMs category is still in silico related study reports. Consequently, these reports would be counted in both categories, potentially inflating the overall numbers, and affecting the accuracy of the data representation.

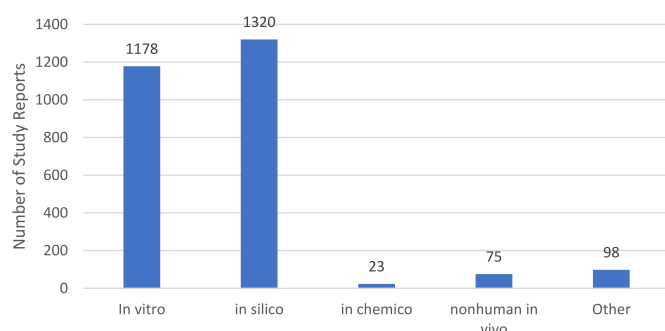


Fig. 1. Total number of Study Reports Identified in INDs, NDAs and BLAs (2010–2024).

Table 2

In vitro related categories and keywords used to search FDA database.

In Vitro NAM Categories	In Vitro Related Keywords	Application Type	
		IND	NDA & BLA
3D Models	3D in vitro	2	0
	3D model	3	0
	3D tissue	1	0
	3D organ	3	0
	3D cell culture	2	0
	3D bioprinting	0	0
	3D tissue printing	0	0
	3D in vitro model	1	0
	3D organoid	1	0
	organoid	36	2
	spheroid	72	6
	tumourid	2	0
	tumouroid	0	0
Coculture or Triculture	assembloid	0	0
	co culture OR co-culture OR coculture	197	23
	tri culture OR tri-culture OR triculture	0	0
Sandwich Culture	sandwich	196	56
	stem cell	305	30
Stem Cell Derived Models	induced pluripotent stem cell	161	3
	OR iPSC		
Reconstructed Tissue Models	reconstructed	41	22
	organ chip	0	0
Organ Chip or MPS Models	Chip (NOT kchip NOT genechip)	4	1
	microphysiological system	2	1
	microfluidic	3	0
	multi-organ	0	2
	multi organ	0	0
TOTAL		1032	146

Data represents number of study reports submitted in an IND, NDA or BLA application for each keyword.

Many sponsors submit numerous study reports under a single application (IND, NDA or BLA) on the same day, indicating that multiple in silico study reports are connected to a single application. For example, in 2024, there were 156 different in silico study reports submitted under one NDA, each representing various impurities tested across different in silico platforms (Fig. 3b). For NDA and BLA applications in 2024, there were 213 study reports that included quantitative structure activity OR QSAR in their study titles, these represented only 12 unique NDA applications (Fig. 3b). There is a similar pattern for INDs. For IND applications in 2024, there were 59 study reports that used quantitative structure activity OR QSAR in the title, however, 50 of the 59 study reports (85 % of the study reports) were attributed to one IND, submitted on the same day (Fig. 3b).

When comparing in vitro and in silico NAMs in regulatory submissions, we observe distinct patterns. In vitro NAMs are used in a greater number of individual IND, NDA and BLA applications while in silico NAMs tend to have more studies submitted per individual application. Again, we must highlight that the numbers of in silico NAMs can be misleading due to overlapping terminology in study titles.

The primary in chemico NAMs submitted in nonclinical packages are direct peptide reactivity assays. These assays evaluate the skin sensitization potential of chemicals, through their covalent binding to proteins, which is a key event of the skin sensitization Adverse Outcome Pathway (AOP) (OECD TG 442). Our database searches only yielded results when using the keywords “direct peptide reactivity assay” and “in chemico” (Table 4). Searches using the keywords “amino acid reactivity assay” or “kinetic direct peptide reactivity assay” did not produce any results (Table 4). All the hits for “in chemico” overlapped with the hits for “direct peptide reactivity assay” which indicates that future searches should combine both keywords (“direct peptide reactivity assay” OR “in chemico”) to ensure a comprehensive list without duplicate study reports (Fig. 4a). The number of study reports related to direct peptide reactivity assays are very limited, with a total of 12 studies submitted (including IND, NDA and BLA submissions) over the past 6 years, despite searching the databases for 15 years (2010–2024) (Fig. 4b).

Nonhuman in vivo NAMs encompass studies performed in alternative organisms including lower organisms such as *C. elegans* and zebrafish (Table 5). Among these, zebrafish studies were more frequently reported (Fig. 5a). The zebrafish studies employed a variety of test methods and endpoints, including assessments of embryotoxicity in embryos, acute toxicity in adult fish, and endocrine activity in embryos. When examining the submission patterns of these studies, there appears to be a relatively consistent distribution of zebrafish studies submitted to IND applications, with approximately 2–6 studies submitted per year (Fig. 5b). In contrast, fewer zebrafish studies were submitted to NDA or BLA applications (Fig. 5b). There are limited assays and submissions for nonhuman in vivo NAMs, with our list of keywords.

The “Other NAMs” category encompasses NAMs that don’t easily fit into the in vitro, in silico, in chemico, or nonhuman in vivo classifications. Our database searches for these NAMs yielded results in three general categories: measurement modalities, bioprinting, and bioengineering (Table 6). Among these, the keywords “multi-electrode array OR multi electrode array OR MEA”, “bioPrint OR bio-print”, and “organo-typic” had the highest number of results (Table 6 and Fig. 6).

Looking at the submission trends over the past 15 years, MEA studies first appeared in CDER submissions in 2014, with a slight increase observed from 2021 to 2023. However, it’s important to note that this recent increase is largely due to multiple submissions under the same IND. For instance, in 2021 and 2023, while there were 11 and 13 IND submissions respectively that mentioned MEA in the title, these represented only 6 and 8 unique INDs respectively. Similar to trends observed with in silico NAMs, a single IND submission often contains multiple MEA study reports. These reports may cover different endpoints or



Fig. 2. In vitro NAMs submitted to CDER
Data reflects the keyword, total number of study reports (INDs, NDAs and BLAs), and percentage within the in vitro NAMs category.

compounds, such as a prodrug and its metabolite. For example, one submission might include studies measuring both the potential to prolong field potential duration and reduce spike amplitude (potentially causing Torsade de Pointes arrhythmias in hiPSC-CMs), as well as studies on neuronal network activity in rat cortical neurons (examining effects on the acute central nervous system).

After analyzing the data for the “Other NAMs” category, the search keywords and submission patterns for these miscellaneous NAMs did not reveal any other obvious or remarkable insights beyond what has been previously discussed.

Table 3
In Silico Related Categories and Keywords Used to Search FDA databases.

In Silico Categories	Computational Methods and Models Related Keywords	Application Type	
		IND	NDA & BLA
In silico	in silico	592	279
	quantitative structure activity OR QSAR	163	273
Computational method	computational method OR computational model	3	0
Other Related Terms	machine learning	0	0
	artificial intelligence	0	0
	predictive model	1	0
	high throughput screen	3	1
	quantitative structure toxicity OR QSTR	0	0
	read across OR read-across	2	3
TOTAL		764	556

Data represents number of study reports submitted in an IND, NDA or BLA application for each keyword.

4. Discussion

There has been a lot of interest regarding the amount and types of NAMs submitted to FDA in drug applications. For this reason, we conducted the present analysis which shows preliminary results providing a high-level overview of the prevalence and types of NAMs used in drug development submissions to the FDA over the past 15 years. It's important to understand that these results are raw and unrefined and intended to provide a landscape overview of NAMs in CDER regulatory submissions. We have not yet removed duplicate entries from the dataset, nor have we manually reviewed individual study reports to confirm their accuracy to the specific NAM categories to which they were assigned here. The analysis did not evaluate the contexts of use, methods, endpoints, or conclusions from the models used in the study reports. For example, the search results for stem cell derived models included a variety of study reports, such as embryotoxicity screening and cardiac toxicity potential assessment, highlighting the broad utility of these models in drug development and safety evaluation. This also highlights how context of use is very important, when using such broadly applicable models. We acknowledge the need for further exploration and analysis of the results to remove duplicates and to establish whether sponsors included the rationale for including specific in vitro NAMs in nonclinical safety packages. However, this preliminary analysis provides a high-level overview which is very informative. Future publications will build upon this work through enhanced data refinement and validation to enable more precise analyses. While our analysis has some limitations, the data still provide valuable initial insights into the usage of NAMs in drug development submissions, such as in vitro assays are the most diverse and common types of NAMs

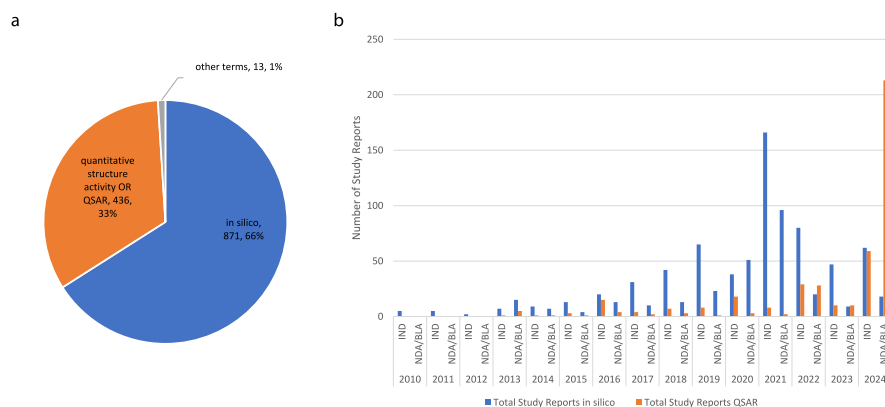


Fig. 3. In silico NAMs submitted to CDER

Data reflects the keyword, total number of study reports (INDs, NDAs and BLAs), and percentage within the in silico NAMs category.

Table 4

In Chemico Keywords Used to Search FDA databases.

In Chemico Related Keywords	Application Type	
	IND	NDA & BLA
direct peptide reactivity assay	10	2
Amino Acid Derivative Reactivity Assay	0	0
kinetic Direct Peptide Reactivity Assay	0	0
In chemico	9	2
TOTAL	19	4

Data represents number of study reports submitted in an IND, NDA or BLA application for each keyword.

submitted in drug applications, while in silico methods are used for limited situations, but can often be submitted in large numbers in very few submissions. This high-level view serves as a starting point for understanding trends in NAM adoption but should be interpreted with certain caveats in mind.

The data confirm that the FDA has been receiving and reviewing NAM data in drug submissions, even before the statutory language updates of FDORA. Overall trends in NAMs submissions indicate that 93 % of NAMs fall into two categories, namely in vitro and in silico. While it is too early to make any assessment, we do not see any significant increases in the submission of NAMs since 2022, when FDORA was passed. Within the in vitro category some NAMs are more developed, and Sponsors may have more confidence in them. This is supported by the data showing increased submissions of data using stem cell derived models, sandwich and coculture models in IND, NDA and BLA applications. The lower rates of submissions for 3D models, MPS and organ chips indicate that despite increased interest in these complex in vitro models in drug discovery there is a reluctance to submit these NAMs data in drug development

Table 5

Nonhuman in vivo Related Keywords Used to Search FDA Databases.

Nonhuman in vivo Related Keywords	Application Type	
	IND	NDA & BLA
<i>C. elegans</i>	0	1
zebrafish	59	15
TOTAL	59	16

Data represents number of study reports submitted in an IND, NDA or BLA application for each keyword.

packages to FDA. The potential barriers to NAM submission in drug development packages may include lack of confidence in study validation and uncertainty about regulatory value and interpretation.

5. Conclusion

CDER is committed to increasingly integrating the use of NAMs in regulatory decision making and drug development. This can be achieved through increased confidence in the data obtained from the use of these methods, as a direct result of increased submission of NAMs in drug applications. Using a CDER search tool to analyze NAMs submitted to CDER, this paper provides a starting point for curating study reports and data from a wide variety of NAMs to prioritize resources to identify promising NAMs with the greatest potential for regulatory impact. We encourage sponsors to include relevant keywords in NAM study report titles to facilitate efficient searching and tracking of NAMs so that CDER can more efficiently identify NAMs data, identify trends and develop metrics around their use.

With increased NAMs submissions to CDER in drug development packages, sponsors, and regulators together, can build scientific confidence on the use of NAMs to improve drug development. Increased

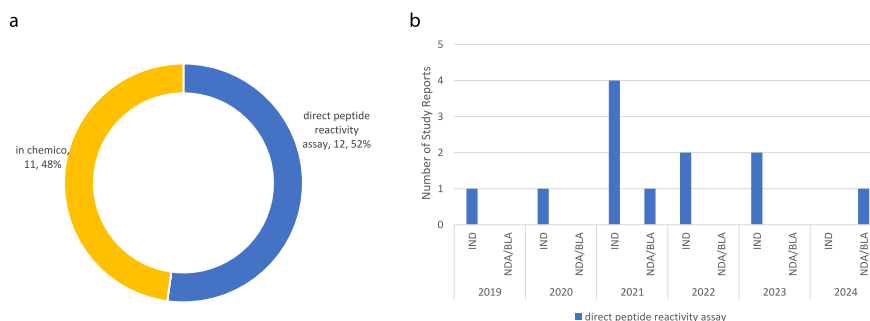


Fig. 4. In chemico NAMs submitted to CDER

Data reflects the keyword, total number of study reports (INDs, NDAs and BLAs), and percentage within the in chemico NAMs category.

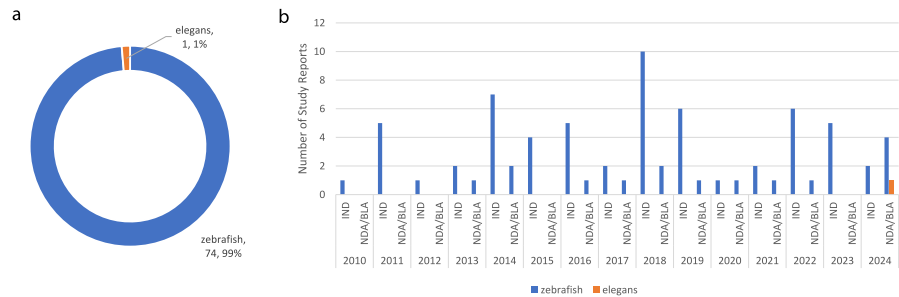


Fig. 5. Nonhuman in vivo NAMs submitted to CDER
Data reflects the keyword, total number of study reports (INDs, NDAs and BLAs), and percentage within the nonhuman in vivo NAMs category.

Table 6
Other NAMs categories and keywords used to search FDA databases.

Other NAMs Categories	Other NAMs Related Keywords	Application Type	
		IND	NDA & BLA
Measurement Modalities	multi-electrode array OR multi electrode array OR MEA (NOT measured NOT measure NOT measuring NOT permeability NOT permeation)	41	0
Bioprinting	TEER OR trans epithelial electrical resistance (NOT volunteer NOT volunteers)	4	0
	bioPrint OR bio-print	30	2
	Micropattern	1	0
Bioengineering	Biomimetic	4	0
	Organotypic	11	1
	engineered tissue	0	0
	engineer AND tissue OR bioengineer	3	1
TOTAL		94	4

Data represents number of study reports submitted in an IND, NDA or BLA application for each keyword.

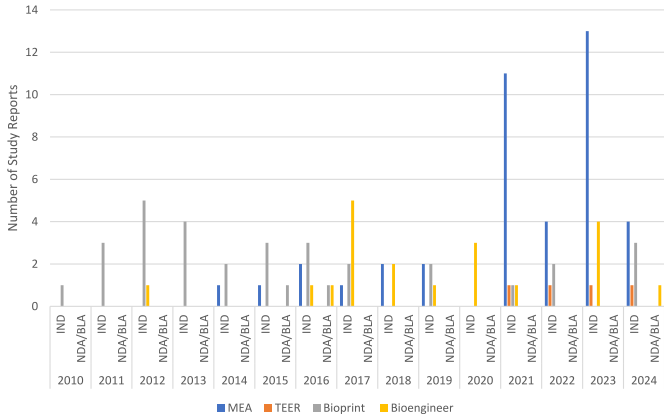


Fig. 6. Other NAMs submitted to CDER
Total Number of Other NAMs Study Reports Submitted by Year
Data reflects both categories and keywords related to Other NAMs.

NAMs submission generates more data on NAMs, which enables CDER to work towards consensus on validation principles, content, and format of NAM study reports. By evaluating increasing amounts of NAM data submitted to CDER in drug development packages, we aim to better meet the FDA's NAM Roadmap objectives.

CRedit authorship contribution statement

Tyna Dao: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation,

Conceptualization. **Nakissa Sadrieh:** Writing – review & editing, Supervision, Formal analysis, Conceptualization.

Disclaimer

None.

Data availability statement

Data for this article are not publicly available due to the confidential nature of regulatory submission data contained within FDA databases. The analysis is based on proprietary information from drug applications (INDs, NDAs, and BLAs) submitted to CDER.

Funding body information – landscape of NAMs

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

None.

Appendix A. KEYWORDS AND Definitions Related to NAMs

Alternative Method	Alternative method generally refers to a testing strategy that refines, reduces, or replaces the use of animal testing to support benefit-risk assessment for FDA-regulated products and differs from CDER's current recommended approach; can include new approach methodologies (NAMs) ^a
Alternative Organism	Alternative organisms are living entities not currently included in the recommended nonclinical studies used to support drug and biologic development for regulatory use
Assembloids	Assembloids consist of self-organizing cellular systems resulting from the combination of a type of organoids with another type of organoids (for example, dorsal forebrain with ventral forebrain) or with different specialized cell types (for example, cortical organoid with endothelial cells) that result in integration ^b
Complex in vitro models (CIVM)	Complex In Vitro Models (CIVM) are systems that have a multicellular environment within a biopolymer or tissue-derived matrix; a 3D structure; and features such as mechanical factors (e.g., stretch, perfusion), primary or stem cell-derived cells, and/or immune system components ^c
Computational simulation	Computational Simulation is the process of representing a real-world system on a computer and then running the simulation by implementing a numerical scheme
Computational toxicology or in silico	Computational (or in silico) toxicology approaches attempt to understand toxicity mechanisms and potential toxic effects using mathematics, informatics, and computer models
Context of use (COU)	Context of use (COU) is a clearly articulated description delineating the manner and purpose of a particular alternative method to address a question of interest for regulatory use ^d
Ex vivo (treated outside of the body)	Refers to experiments done in tissues from an organism in an external environment (outside the living body) with minimal alteration of natural conditions. An example is perfusing an intact liver in the lab to determine how a chemical is processed ^b
Human induced pluripotent stem cells (hiPSC)	Human induced pluripotent stem cells (hiPSC) are a type of pluripotent stem cell, derived from somatic cells of human donors that have been genetically programmed and then differentiated into different types of human tissues or primary cells (e.g. hepatocytes, cardiomyocytes)
In chemico	In Chemico Tests represent abiotic chemical reactivity methods, usually physicochemical measurements that do not contain biological material (with the exception of certain tests that employ enzyme fractions) ^e
In vivo	Refers to experimentation using a whole, living organism ^b
In vitro	In vitro tests refer to the technique of performing a given procedure in a controlled environment outside of a living organism. This usually involves studying isolated cells by themselves or in concert with others ^b
Immortalized cells	They derive from human or animal sources and have been manipulated to proliferate indefinitely in vitro and can thus be cultured for long periods of time. This immortalization can be induced through genetic engineering or can be because of isolation from sources that are chromosomally abnormal or that carry mutations that enable continuous cell division ^f
Microphysiological systems (MPS)	Microphysiological Systems represent a subset of CIVMs and are complex, multicellular in vitro systems that commonly include three-dimensional aspects, fluid flow, changing pressure or stretch, and multi-organ interactions ^b
New approach methodology (NAM)	New approach methodology(ies) include a broad range of methods such as complex in vitro, 2-D in vitro, in chemico, and in silico methods. In vivo methods can also be considered NAMs when they improve predictivity, shift studies to phylogenetically lower animals, or otherwise help replace, reduce, and refine animal use (i.e., the 3Rs) in development programs. Such NAMs can improve regulatory efficiency and potentially expedite drug development ^g
Organ-on-a-chip systems	Organ-on-a-Chip systems, also called tissue chip or organ chip, represent a subset of MPS consisting of a miniaturized physiological environment engineered to yield and/or analyze functional tissue units capable of modeling specified/targeted organ-level responses (e.g., lungs) ^b
Organoids	Organoids are in vitro-generated cellular systems that emerge by self-organization, include multiple cell types, and exhibit some cytoarchitectural and functional features reminiscent of an organ or organ region. Organoids can be generated as 3D cultures or by a combination of 3D and 2D approaches (also known as 2.5D) that can develop and mature over long periods of time (months to years) ^c
Multipotent stem cells	These are undifferentiated cells considered to be able to self-renew and differentiate into all cell types within one lineage (e.g., blood stem cell, mesenchymal stem cell) ^g
Pluripotent stem cells (PSC)	PSCs are undifferentiated cells of embryonic or somatic origin that can differentiate into cells of the three embryonic germ layers (ectoderm, mesoderm, and endoderm) and have self-renewing capacity. There are various states of pluripotency that resemble different states of the embryo, not just one. It is important to demonstrate what qualitative and quantitative assays have been performed along with the markers that are used to demonstrate the state of pluripotency of the cells
Primary cells	These are freshly isolated directly from living organism and maintained from growth in vitro. They enable researchers to study the morphological, cellular and functional behavior of the tissue ^g
Spheroid	Spheroids, a type of 3D cell culture, consists of spherical cellular units that are generally cultured as free-floating aggregates and are of lower complexity structurally, compared to organoids, but are simple and popular models for drug screening ^h
Qualification	Qualification is a process that allows for an alternative method to be endorsed by FDA in advance for a specific COU. Qualification is a conclusion that within the stated context of use, an alternative method can be relied upon to have a specific interpretation and application in regulatory review ^b
Validation	Validation is a process by which the accuracy, reliability, and relevance of a procedure are established for a specific COU. Validation for one context of use does not imply validation for other COUs ^e
3Rs	3Rs are the principles of reducing (e.g., minimizing the number of animals used), replacing (e.g., use of a lower-order species or non-animal test), and refining (e.g., updating a test methodology to better inform risk) the use of animals in drug development
Totality of evidence	Totality of evidence refers to the sum of all available data intended to inform a regulatory decision
Weight of evidence approach	A Weight of evidence approach is an integrated assessment used to determine the overall risks associated with any safety signals taking into consideration all available data, whether from alternative methods or in vivo animal toxicity studies, and considering the relevance of each model to the intended clinical use ⁱ

Sources where definitions and keywords were taken or adapted from.

^a FDA AM Website 2023 About Alternative Methods (FDA) (2023) <https://www.fda.gov/science-research/advancing-alternative-methods-fda/about-alternative-methods>.

^b NIH NAM Report: Catalyzing the Development and Use of Novel Alternative Methods Catalyzing the Development and Use of Novel Alternative Methods (2023).

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Data availability

The data that has been used is confidential.

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